
Thermal Methods

Active Thermal Testing of Delaminations in Heat-Shielding Structures

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Abstract—We describe possibilities offered by active thermal testing when detecting delaminations of heat-shielding coatings from a metal base using an optical thermal-stimulation source that implements heating in a scanning mode. Results of numerical simulation of thermal processes are given for various defective situations. The thermophysical characteristics of heat-shielding materials have been experimentally determined, and the limits of the thermal method have been estimated for heat-shielding thickness and defect size.

Keywords: active thermal testing, heat-shielding coating, defect, delamination, optical heating

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INTRODUCTION

In the context of introducing new materials and technologies in aviation and rocket and space technology and considering a number of emergencies and accidents of recent years, the development of effective methods and equipment for nondestructive testing (NDT) in these industries is an important task. The areas of individual objects to be tested can reach 300 m²; this places increased demands not only on the technical parameters of the applied method of testing but also on its performance [1–3]. High NDT performance can be achieved using the visual-measuring method, which does not reveal hidden defects. Due to the poor productivity of testing, traditional methods (in particular, ultrasonic and radiation ones) are mainly used for local diagnostics. In addition, the use of these methods for the NDT of heat-shielding coatings of rocket and space engineering products encounters fundamental difficulties due to the low attenuation of ionizing radiation and the substantial damping of ultrasonic waves. Active infrared (IR) thermography (thermal testing) is a noncontact method that does not require two-sided access and provides high testing performance, sufficient reliability, and the possibility for automating the testing with robots [4–6]. Nevertheless, due to both objective and subjective reasons, thermal testing remains a qualitative (screening) technique that needs to be supplemented with NDT methods that are more on the “defect-characterization” side. In the present work, one-sided thermal testing of delamination of heat-shielding coatings from the metal base has been analyzed, with the testing performed on the coating side.

TEST OBJECT

The procedure of active thermal testing was developed and its optimum parameters were determined using a two-layer sample with dimensions 250 × 190 × 4 mm, which was a foamy heat protection applied to a 4-mm-thick duralumin base. The heat-protection thickness was 2.8, 5.2, and 7.2 mm (Fig. 1). Characteristic defects in such structures are delaminations of the heat-insulating layer from the metal, as well as delaminations between successively applied layers of thermal insulation. It should be noted that applying thermal insulation to a metal base is a technologically complex procedure, and creating artificial delaminations of required dimensions in it is difficult. The often used imitation of delaminations with polymethyl methacrylate inserts can give rise to unplanned (extra) defects. In the investigated sample, defects were introduced by removing the adhesive layer from the duralumin substrate before applying thermal-insulation coating. The two-layer sample contained nine defects of delamination of the heat-

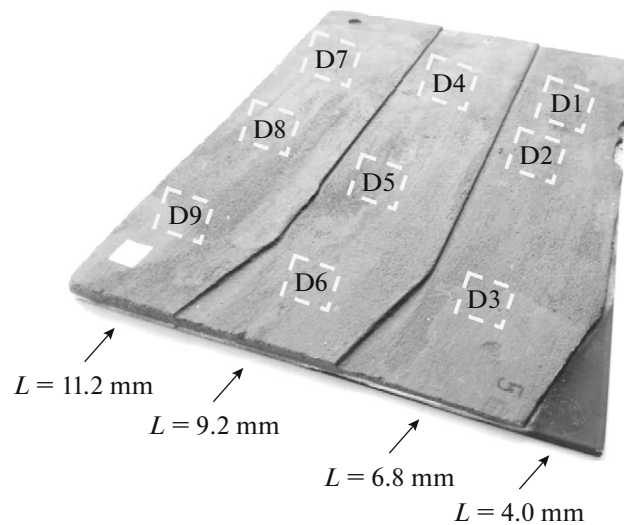


Fig. 1. Sample of heat-shielding coating on duralumin base.

insulating layer from the substrate with an opening of 0.2 ± 0.1 mm and dimensions 10×10 , 15×15 , and 20×20 mm (see the diagram of defects in Fig. 1).

THREE-DIMENSIONAL MODELING OF ACTIVE THERMAL TESTING

Detecting delaminations between the metal base and the heat-shielding layer, depending on the thickness of the latter, requires different heating and observation times, reaching several tens of seconds. The magnitude of the temperature signal over defective area is directly proportional to the heating power; however, the quality testing of applying a heat-shielding layer on certain types of products at the final stages of their production is limited by the allowable surface heating temperature, in many cases equal to $+60^\circ\text{C}$. Thus, at the ambient temperature of $+20^\circ\text{C}$, the maximum excess surface temperature should not exceed 40°C , imposing a limitation on the heating power.

Three-dimensional modeling was performed to determine the optimum parameters of thermal testing, namely, the duration of heating and the time of detecting delaminations buried under different thicknesses of thermal protection. The ThermoCalc-3D (Tomsk Polytechnic University) software was used. It implements typical models of active thermal testing (Fig. 2) [7].

At the first stage of the research, the temperature at the surfaces of the two-layer duralumin–heat-shielding structure was analyzed with one- and two-sided active thermal testing (Figs. 2–4). A two-layer

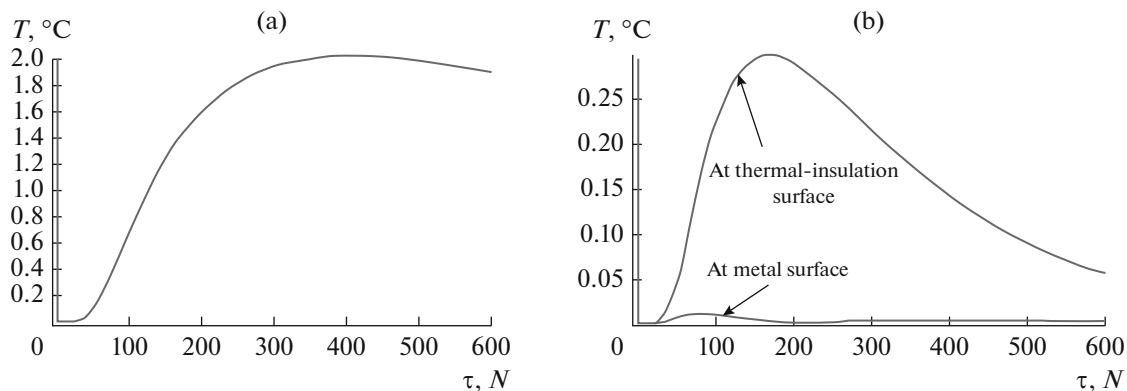


Fig. 2. Change in temperature over time under two-sided thermal testing of the two-layer structure of duralumin (4 mm) and heat protection (7.2 mm) when heated on the metal and on the heat-protection side: (a) graphs for defect-free region (temperature curves coincide); (b) temperature signals for defective areas.

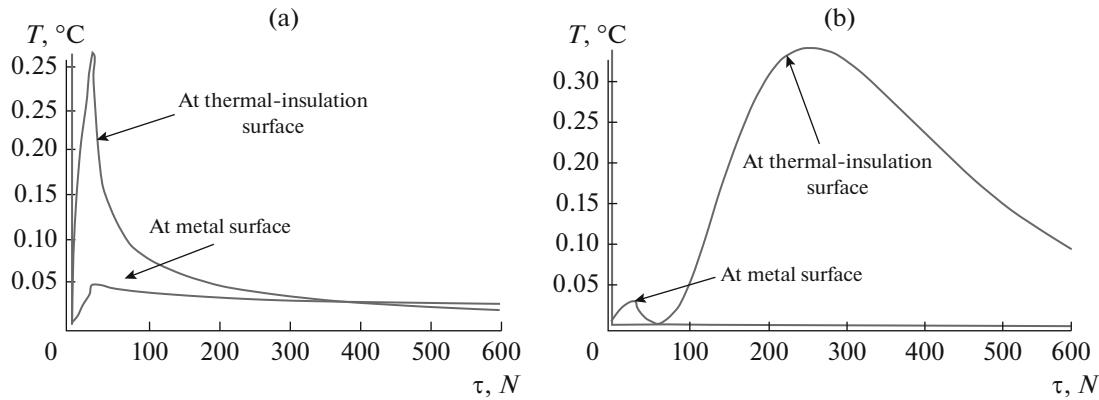


Fig. 3. Simulation results for one-sided thermal testing of two-layer structure: (a) graphs of temperature changes for defect-free areas; (b) temperature signal in defective area.

structure with the thickness of 11.2 mm was considered: duralumin (4 mm) plus heat insulation (7.2 mm) with a defect between them in the form of an air cavity with a size of $D = 20 \times 20$ mm and an opening of $h_d = 0.2$ mm. The heating time was 15 s with a heating power density of 3400 W/m^2 achieved with two halogen lamps (1 kW each) used in the experiments.

The thermophysical characteristics of materials used in the simulation are given in Table 1. The reference data [8, 9] were used for duralumin and air, and the thermophysical characteristics of thermal insulation were found by applying a modified Parker method to testing the 2.8-mm-thick thermal-protection sample [10,

11]. The thermal diffusivity a was determined experimentally using the formula $a = \frac{0.139L^2}{\tau_{0.5} - 0.5\tau_h}$, where L is

the sample thickness (2.8 mm), $\tau_{0.5}$ is the time of reaching half the maximum signal at the rear sample surface (2.9 s), and τ_h is the duration of heating (5 s). The thermal conductivity of the thermal protection was calculated by the formula $\lambda = W a \tau \varepsilon / T L$ [10], where W is the heating power density, equal to 3400 W/m^2 ; $\tau_h = 5$ s; ε is the absorption coefficient (0.95); and T is the maximum increment of the sample temperature (6.13°C). The heating power density was evaluated using a blackened copper plate with known thermophysical characteristics, which was tested under identical conditions. The heat capacity of the heat-shielding layer is $C = \lambda / (a\rho)$ [10], with the density of the heat-shielding layer determined by recalculating the mass and overall dimensions of the sample obtained by direct measurements. The error in the above estimates of thermophysical characteristics was estimated at 15–20% by applying the two-sided thermal-testing procedure (the Parker method) to testing blackened samples of various materials (copper, aluminum, and polymethyl methacrylate) with known thermophysical characteristics.

Considering the fact that thermal testing is considered as a qualitative method of nondestructive testing, the above-mentioned error in determining the main indicators of thermal-testing procedures is acceptable.

The above values of thermophysical characteristics were used in the simulation of defective situations. Under two-sided thermal testing, the order of arranging sample layers relative to the source of heating and the thermal imager does not affect the nature of temperature curves in defect-free areas (Fig. 2a); however, significant differences are observed in defective zones (Fig. 2b). This is due to the fact that temperature sig-

Table 1. Thermophysical characteristics of materials used in modeling [8, 9].

Material	Heat conductivity, $\text{W}/(\text{m K})$	Density, kg/m^3	Heat capacity, $\text{J}/(\text{kg K})$	Thermal diffusivity, m^2/s
Duralumin	177	2770	875	73×10^{-6}
Air	0.02	1.10	1030	1.90×10^{-5}
Heat protection*	0.19	470	2000	2.02×10^{-7}

* Determined experimentally.

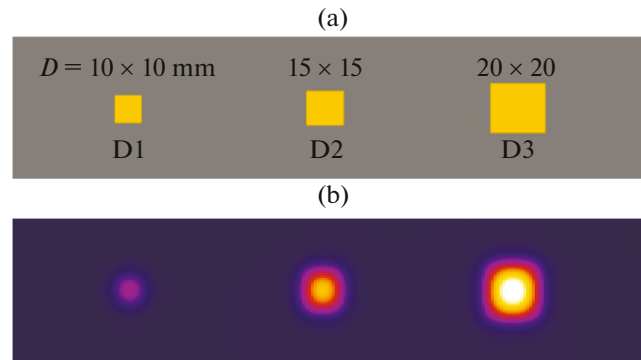


Fig. 4. Three-dimensional model of duralumin–thermal-protection structure: (a) defect scheme; (b) IR thermogram.

nals better survive in the thermal-protection material and quickly smoothen out in the metal, so under heating on the thermal-protection side, defects are not detected on the opposite side of the structure.

It can be seen from the graphs in Fig. 2b that the maximum of the temperature signal $\Delta T_{\max} = 0.001^{\circ}\text{C}$ when testing for delaminations from the metal falls on the point of time $\tau = 42$ s ($N = 84$, where N is the thermogram number and the thermogram recording interval is 0.5 s), while when recording the temperature on the thermal-protection side, the maximum signal value is $\Delta T_{\max} = 0.31^{\circ}\text{C}$ at $\tau = 88$ s ($N = 176$).

The main focus in the modeling was on the practically important procedure of one-sided testing (Fig. 3). The temperature signal in the defective area when heating on the duralumin side was 0.031°C at $\tau = 15$ s ($N = 30$) and on the heat-protection side it was $\Delta T_{\max} = 0.36^{\circ}\text{C}$ at $\tau = 130$ s ($N = 260$).

At the second stage, three numerical models of the product were investigated (see Fig. 1), each containing three defects in the form of air cavities with lateral dimensions $D = 10 \times 10$, 15×15 , and 20×20 mm and thickness $h_d = 0.2$ mm. Defects D1–D3 were located between the duralumin base 4 mm thick and the thermal-protection layer with thickness $L = 2.8$ mm (model no. 1); defects D4–D6 were located at a depth of $L = 5.1$ mm (model no. 2); and defects D7–D9 were buried to a depth of $L = 7.2$ mm (model no. 3). The layout of defects is exemplified with model no. 1 in Fig. 4.

Simulation results are compiled in Table 2. The design parameters were the values of temperature signals $\Delta T = T_d - T_{nd}$ and the running temperature contrast $\text{Con} = \frac{\Delta T}{T_{nd}}$, as well as the times of appearance of their maximum values τ_m .

It can be seen that the value of ΔT for the delamination with a thickness of 0.2 mm and dimensions of 20×20 mm decreases from 4.20 to 0.44°C with increasing the thickness of the heat-shielding layer, and the contrast Con drops from 81 to 15%. The optimum time τ_m for observation of deep defects based on temperature signals is 130 s, reaching 170 s when based on temperature contrasts.

According to the theory of thermal testing, all these quantities can be measured experimentally. In particular, the standard temperature sensitivity of state-of-the-art thermal imagers reaches 0.04°C , and noise contrasts on homogeneous and “blackened” surfaces are experimentally evaluated at the level of 4–5%.

Table 2. Results of numerical simulation.

	Model no. 1 (thickness of thermal-insulation layer 2.8 mm)			Model no. 2 (5.2 mm)			Model no. 3 (7.2 mm)		
h_d , mm	0.2								
D , mm	20×20	15×15	10×10	20×20	15×15	10×10	20×20	15×15	10×10
ΔT , °C	4.20	4.02	3.38	1.05	0.95	0.63	0.44	0.33	0.19
$\tau_m(\Delta T)$, s	35	34	32	80	77	70	130	123	114
Con, %	81	74	56	28	24	15	15	11	6
$\tau_m(\text{Con})$, s	48	47	44	108	103	94	170	162	150

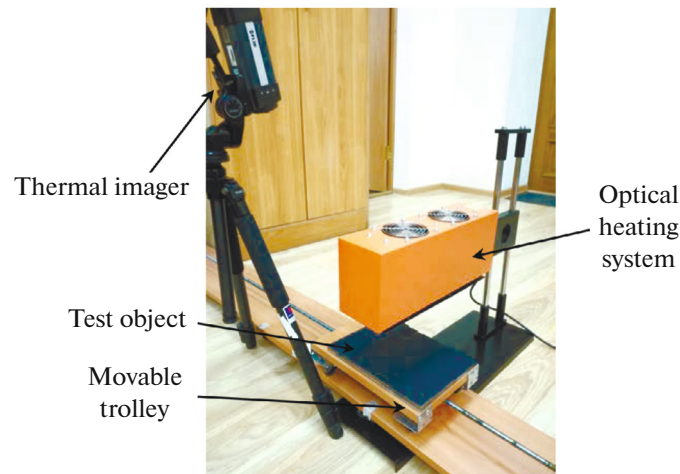


Fig. 5. Experimental setup for thermal testing with a linear heating source in scanning mode.

EXPERIMENTAL RESULTS OF ONE-SIDED THERMAL TESTING

Experimental studies were performed using the method of scanning with a linear heat source (see the experimental setup in Fig. 5).

The main parameters of experimental procedure (the power and duration of heating, thermogram-recording frequency) were chosen according to the simulation results. The sample was mounted on a movable trolley, driven by a computer-controlled engine, and displaced through a strip heating zone created by linear halogen lamps with a combined power of 2 kW. The product displacement speed was chosen equal to 6.5 mm/s, which, with a heating zone width of 100 mm, provided heating each product point for 15 s. The temperature at the surface of the sample after it had left the heating zone and stopped was recorded with a FLIR A325 thermal imager for 200 s at an image recording frequency of 2 Hz. Figure 6 shows thermograms for different observation times that correspond to the appearance of maximum temperature signals ΔT from delaminations situated beneath various heat-protection thicknesses.

The low contrast of defect indications in Fig. 6 is primarily associated with the small thickness of defects h_d and a considerable thickness of thermal protection. Note that extensive irregularly shaped delaminations were discovered in the zone of defects D1, D6, D7, D8, and D9. Defect D5 was not detected, apparently, due to the absence of delamination. Table 3 lists the experimental values of temperature signals ΔT and temperature contrasts Con , as well as the optimum observation times $\tau_m(\Delta T)$ and $\tau_m(\text{Con})$. It can be seen that the values of temperature signals and contrasts for defects located beneath

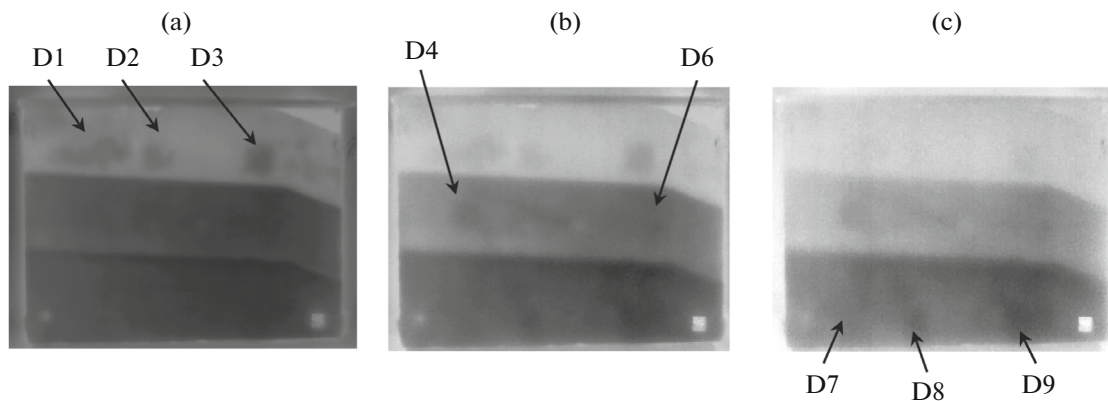


Fig. 6. Thermograms of thermal-protection–duralumin structure for different observation times (the palette is inverted) obtained with one-sided thermal testing: (a) observation time 34 s (optimum for thermal protection 2.8 mm thick); (b) 101 s (optimum for 5.2 mm); (c) 151 s (optimum for 7.2 mm).

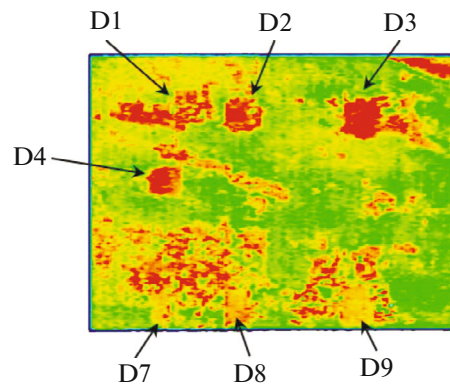


Fig. 7. Result of ultrasonic testing: C-cut at depth 4 mm.

the insulation layer with the thickness of 7.2 mm are at the limit of sensitivity and “fade” into temperature noise.

To verify the results of thermal testing, the tested sample was inspected by an ultrasonic method using a focused immersion sensor operating at a frequency of 15 MHz with full immersion of the product into the immersion liquid. Scanning was carried out on the metal side. The results are depicted in Fig. 7. The red color corresponds to zones where the contact between the thermal protection and the base is violated. Regularly shaped domains were discovered that correspond to defects D1, D2, D3, D4, D7, D8, and D9.

After completion of the experiments, the heat shield was removed in the zone of defects D1–D2 (Fig. 8). An area of violation of contact between the thermal protection with the metal base was discovered around defect D1. This area was characterized by an extensive temperature anomaly according to the results of thermal testing and by a delamination as per the results of ultrasonic testing (see the zone of defect D1 in Figs. 6a and 7).

LIMITS OF ONE-SIDED THERMAL TESTING

Figure 9 presents a diagram illustrating the possibilities of active thermal testing for delaminations of thermal protection from metal base. The diagram is based on the results of modeling one-sided thermal testing and shows dependence of temperature contrasts on the size of defects and their depth. The model parameters corresponded to the thermophysical characteristics of the examined standard sample shown in Fig. 1; the parameters of the source of heating corresponded to halogen lamps with the total power of 2 kW. A total of 128 defective situations were calculated, in which delaminations with thicknesses $h_d = 0.2, 0.4, 0.6$, and 0.8 mm and dimensions $D = 20 \times 20, 15 \times 15, 10 \times 10$, and 5×5 mm were placed between the 4-mm-thick duralumin base and the heat-shielding layer with thickness $L = 2, 4, 6, 8, 10, 12, 14, 16$, and 18 mm. The limits of thermal testing were estimated on the basis of the noise contrast $Con_n = \sigma_{nd}/T_{nd}$, experimentally determined at the surface of the heat-shielding layer, where σ_{nd} is the standard deviation

Table 3. Experimental thermal-testing results.

	Heat-insulating layer thickness 2.8 mm			Heat-insulating layer thickness 5.2 mm			Heat-insulating layer thickness 7.2 mm		
h_d , mm	0.2								
Defect no	D1	D2	D3	D4	D5	D6	D7	D8	D9
D , mm	10×10	15×15	20×20	10×10	15×15	20×20	10×10	15×15	20×20
ΔT , °C	1.51	1.41	2.59	0.46	—	0.27	0.09	0.06	0.25
$\tau_m(\Delta T)$, s	40	40	38	96	—	101	166	152	160
Con, %	6	5	9	2	—	1	0.4	0.2	1
$\tau_m(\text{Con})$, s	40	40	40	100	—	101	166	152	160

* The maximum excess temperature of surface heating did not exceed +31°C.

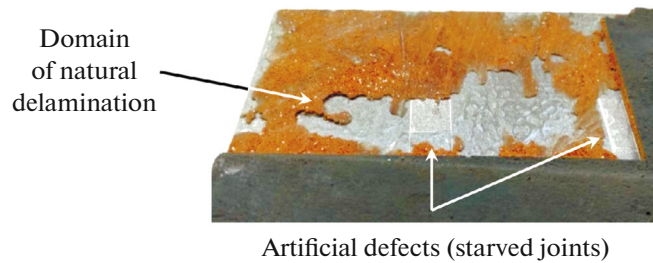


Fig. 8. Zone of defects D1–D2 after removal of heat-shielding layer.

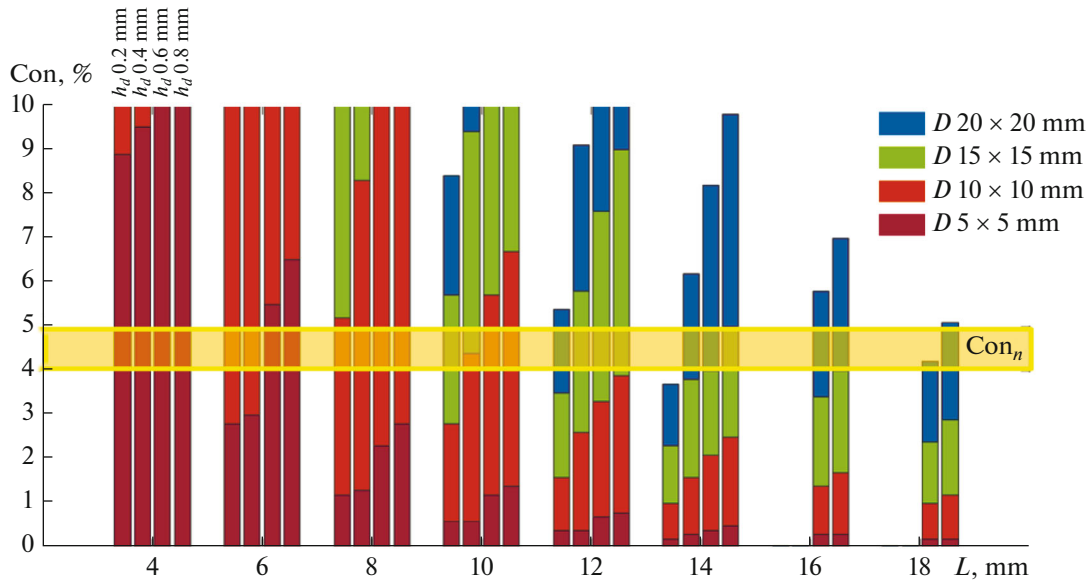


Fig. 9. Possibilities of active thermal testing when detecting delaminations of thermal protection from metal base: the red, orange, green, and blue colors of columns refer to temperature contrasts of defects with sizes $D = 5 \times 5$, $D = 10 \times 10$, $D = 15 \times 15$, and $D = 20 \times 20$ mm, respectively. Four columns for each value of L show the contrast for defect thickness $h_d = 0.2, 0.4, 0.6$, and 0.8 mm (left to right).

of noise signals in the defect-free area and T_{nd} is the average excess temperature within the same area. Depending on the choice of the defect-free zone at the surface of the standard sample, the experimental value of Con_n varied from 4 to 5% (shown in Fig. 9 as the yellow horizontal band).

It can be seen from the diagram that, for example, a defect with size 5×5 mm and thickness 0.2 mm can be detected underneath the heat protection with a thickness of no more than 4 mm. The same defects with thicknesses of 0.6 and 0.8 mm can be detected underneath the heat protection 6 mm thick. In general, defects with the thickness of 0.2 mm cannot be detected for the thermal-protection thickness of more than 12 mm. Note that thermal protection with a thickness of 18 mm and more is the limiting thickness for the one-sided procedure of active thermal testing (for h_d less than 0.8 mm). At the same time, the time of development of the maximum temperature contrast from a defect located at the depth of 18 mm can reach 1000 s.

CONCLUSIONS

In the present work, based on numerical modeling and experimental studies, a method is described for estimating the limits of active thermal testing when detecting delaminations of heat-shielding coating from duralumin base. The thermophysical properties of thermal protection are determined experimentally by the Parker method. It is shown that testing on the side of duralumin is inexpedient due to its high thermal

conductivity, while one- and two-sided thermal testing on the side of thermal protection provides close values of temperature signals. The parameters affecting detection are the lateral dimensions and thicknesses of defects, as well as the level of noise at the thermal-protection surface (4–5%). When using halogen lamps as a heat source and for the thickness of delaminations of up to 0.2 mm in the practically important thermal-protection thickness range of up to 7 mm, temperature signals range from 0.1 to 2.6°C, while the optimum recording time is from 40 to 170 s. Considering that the real sensitivity of thermal testing is limited by the temperature resolution of state-of-the-art thermal imagers (0.01–0.04°C), the maximum thickness of thermal protection in the one-sided thermal testing procedure is estimated at 18 mm for delaminations with dimensions 20 × 20 mm and thickness 0.8 mm.

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